Instructions:

- This is an optional extra credit assignment. It will be weighted as an additional problem set in the final grade if completed.
- Submissions are due by 9:00 p.m. Thursday 7/23/2026. Solutions will be released at this tiime.
- We strongly recommend attempting these problems alone first to help gauge your level of understanding for review.

Problem 1. 10 pts: In class we discussed four categories of focusing: quadrupole (strong) focusing, solenoid focusing, bending dipole focusing, and dipole edge focusing.

a) Discuss the differences between solenoid and quadrupole focusing. What is achieved in each case? When might each be used?

b) Assume you have a circular accelerator in the horizontal plane with a mixture of rectangular and sector bend dipoles. In which magnets do the two types of dipole focusing occur, and which transverse dimension do they affect?

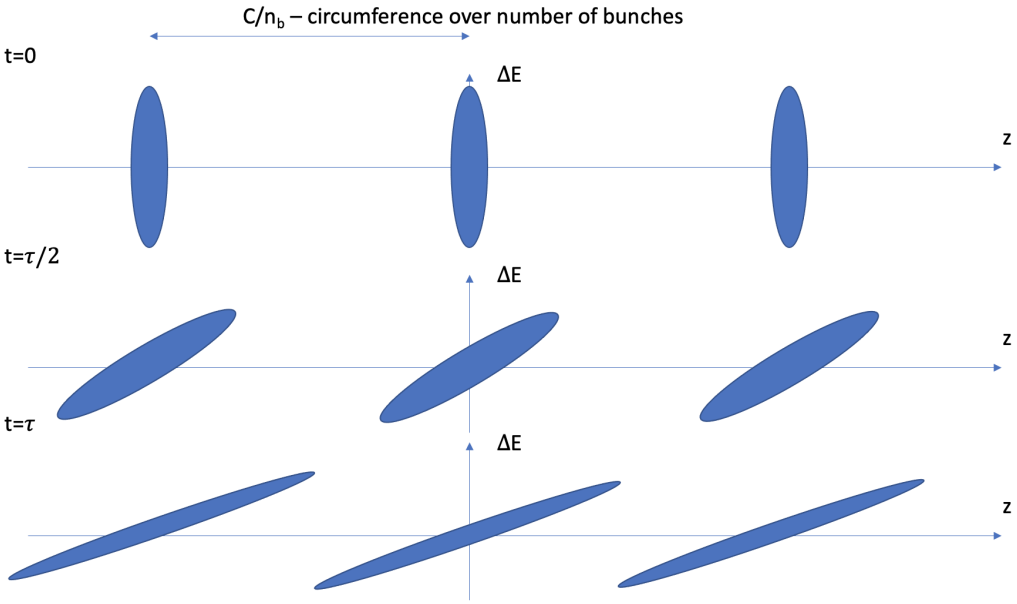
a) Quadrupole focusing uses a higher order multipole field that has a gradient through the entire transverse plane. This gradient causes focusing. Quadrupole magnets inherently focus in one Cartesian transverse plane and defocus in the other due to the shape of the quadrupole field.

In solenoid focusing, the focusing effect occurs from the edge fields which have a gradient. The solenoid magnet is inherently cylindrical and the gradient at the edges focuses in both planes. The fields inside the magnet itself don't focus or defocus. All of the focusing effect comes from the edges.

Solenoid focusing is used more often in low energy cases like particle sources/early stage linacs. Because it relies on the edge effects of a field rather than a higher order multipole gradient, it can be difficult to achieve the same strong focusing as quadrupole magnets. Solenoids also couple the two transverse coordinates, though in the ideal thin lens case the coupling is neglected. Quadrupole magnets are better for high energy particles like those in colliders, synchrotrons, and storage rings. They must be used in a series of focusing and defocusing quadrupoles to achieve stable motion. To first order, they are linear and keep the two transverse coordinates uncoupled.

b) Sector bend dipoles are responsible for bending dipole focusing in the horizontal plane. This is due to the longer path length of particles with larger x. In rectangular dipoles, particles enter at an angle with respect to the pole face. This causes a gradient which results in vertical edge focusing and horizontal edge defocusing. The path length of all particles is the same, so there is no bending focusing in the horizontal plane.

Problem 2. 10 pts: Recall Problem 3 from the midterm. Assume the bunch that is injected has an elliptical phase space with no average correlation in time and energy. Sketch the phase at injection, $\tau/2$, and τ for two adjacent bunches where τ is the debunching time.



Problem 3. 10 pts: In class, we have discussed the Courant-Snyder Invariant

$$C\left(u,u'\right)=\frac{1}{\beta}\left[u^2+\left(\alpha u+\beta u'\right)^2\right]=\gamma u^2+2\alpha uu'+\beta u'^2$$

where u is an arbitrary transverse coordinate.

a) Explain in words the meaning of the Courant-Snyder Invariant. When is the is the Courant-Snyder Invariant actually invariant?

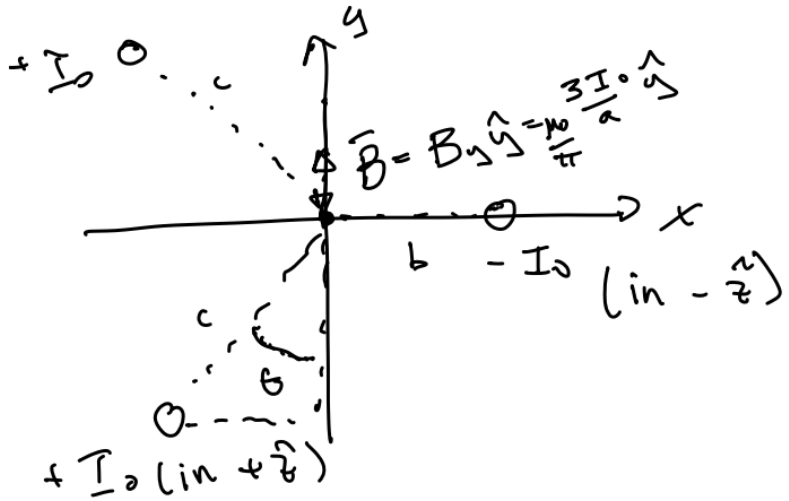
b) Briefly explain the difference between emittance, RMS emittance, and normalized emittance.

a) The Courant-Snyder Invariant describes the ellipse in transverse phase space of a given particle. A particle follows this ellipse as it undergoes betatron oscillations, with some fractional number of betatron oscillations per revolution around the accelerator. The Courant-Synder Invariant is only invariant in the absence of acceleration.

b) The Courant-Snyder Invariant is the emittance ϵ, where the area of the phase space ellipse is $\pi\epsilon$. RMS (root mean square) emittance is based on a statistical distribution of the particles in phase space. It uses the beam widths σ_u, $\sigma_{u'}$, and $\sigma_{uu'}$ to describe the emittance. It can be thought of as the emittance of the RMS particle. Normalized emittance accounts for the change in emittance/the Courant-Snyder Invariant with acceleration. It includes the particle energy/velocity to define an emittance that is invariant with acceleration.

Problem 4. 10 pts: Use three wires to create a field at x=0 and y=0 that only produces magnetic field in the $\hat{y}$ direction, and the magnitude of B_y is $3\mu_0 I_0/(\pi a)$. The wires are carrying a current I_0 each. Keep the wires far apart and the wires cannot overlap.

$$|B| = \frac{\mu_0 I}{2\pi r} \qquad \vec{B} = \frac{\mu_0 I}{2\pi r}\hat{\phi}$$



$$\vec{B} = +\frac{\mu I_0}{2\pi b}\hat{y} + \cos\theta \frac{\mu I_0}{2\pi c}\hat{y}$$

$$-\sin\theta \frac{\mu_0 I_0}{2\pi c}\hat{x} + \cos\theta \frac{\mu_0 I_0}{2\pi c}\hat{y}$$

$$+\sin\theta \frac{\mu_0 I_0}{2\pi c} = \frac{3 I_0 \mu_0}{\pi a}\hat{y}$$

$$\frac{1}{2b}\hat{y} + \frac{\cos\theta}{c}\hat{y} = \frac{3}{a}\hat{y}$$

pick a θ solve for c and a b

Problem 5. 10 pts: Circular Colliders

a) If you want to reduce the radiation losses (power) at 100 GeV for LEP by a factor of 10 how big should the e^+e^- bending radius be?

Increases by $\sqrt{10}$ $\left(P \propto 1/\rho^2\right)$ for a single particle or for constant charge.

The energy loss per turn for a particle is $U = C_\gamma E^4/\rho$. To get the power $P = UI$. If you keep I constant the total charge stored in the ring is increased.

b) What would happen if you went from electrons to muons and kept the same kinetic energy?

Muons are heavier (105.7 MeV vs 0.511 MeV for electrons). We will need a stronger magnet for the same energy particle but there will be less synchrotron radiation ($2x10^9$ less power radiated).